

Paper Auto-Trading System

Following Big Money Flow in

Indonesia Stock Exchange (IDX)

A Technical Architecture Guide

Executive Summary

This document outlines the complete architecture required to build an automated paper trading system that detects and follows institutional ('big money') capital flows within the Indonesia Stock Exchange (IDX). The system is designed to identify accumulation patterns, execute simulated trades, and validate strategies without real capital at risk.

1. Core System Architecture

The system is divided into five interconnected layers, each responsible for a critical function in the trading pipeline.

Layer	Function	Key Components
Data Infrastructure	Ingest & normalize market data	Real-time feeds, historical DB, ownership data
Detection Engine	Identify big money signals	Volume anomalies, broker clustering, foreign flow
Broker Integration	Connect to execution venues	REST/WebSocket APIs, paper trading mode
Execution Engine	Route & manage orders	Order types, position sizing, slippage models
Risk & Compliance	Protect capital & obey regulations	Stop-loss, exposure limits, OJK compliance

2. Data Infrastructure

Reliable data is the foundation of any quantitative system. For Indonesia specifically, the following data streams are essential:

2.1 Real-Time Market Data

- **Price & Volume Ticks:** Best bid/ask, last price, traded volume per tick
- **Order Book Depth:** Level 2 data to detect iceberg orders and walls
- **Transaction Data:** Individual trade prints with size and broker codes
- **Index & Sector Data:** IHSG, LQ45, sectoral indices for relative strength

2.2 Ownership & Flow Data

- **Foreign/Domestic Net Flow:** Daily summaries published by IDX
- **Broker Summary:** Aggregated buy/sell volumes per broker code (via Stockbit/RTI)
- **Block Trade Reports:** Large off-market transactions
- **KSEI Shareholder Data:** Beneficial ownership changes (with delay)

2.3 Data Providers & APIs

Provider	Data Type	Access Method	Cost Level
IDX Direct	Official T+1 data	Website/API	Free / Low
RTI Business	Real-time + broker summary	WebSocket/API	Subscription
Stockbit	Social + broker data	API	Subscription
Valbury	Broker-integrated feed	Broker API	Included with account
Bloomberg/Reuters	Institutional-grade	Terminal	High

3. Big Money Detection Engine

The core intellectual property of the system. This engine translates raw market data into actionable signals by identifying patterns characteristic of institutional accumulation or distribution.

Signal	Definition	Detection Method	Weight
Volume Anomaly	Traded volume >3x 20-day average	Rolling VWAP comparison	High
Price-Volume Divergence	Large volume with minimal price change	Correlation coefficient <0.3 on high volume	Very High
Broker Clustering	Multiple large orders from same broker	Aggregate by broker code, flag top 5%	Medium
Block Trade Alert	Single transaction >IDR 1B or >1% ADV	Fill in tape by size threshold	High
Foreign Flow Spike	Net foreign buy >2x recent average	Daily foreign flow delta	Medium
Order Book Imbalance	Bid/ask ratio >3:1 with increasing depth	Realtime order book analysis	Medium
Opening Auction Surge	Pre-open volume spike with directional bias	Auction phase volume/price analysis	Low

3.1 Composite Signal Scoring

Individual signals are combined into a composite **Big Money Score (BMS)** ranging from 0 to 100. A score above 70 triggers a 'watch' alert; above 85 triggers paper trade entry consideration. Weights are dynamically adjusted based on backtested performance per signal.

3.2 False Signal Mitigation

- **Retail FOMO Filter:** Exclude signals where retail buy ratio >60% of volume
- **News Correlation Check:** Cross-reference with material news to avoid event-driven noise
- **Sector Rotation Context:** Confirm signal aligns with broader sector inflow
- **Time-of-Day Adjustment:** Weight opening session signals lower due to volatility

4. Broker API Integration

Paper trading requires a broker that supports API access with a simulated (paper) account. The following Indonesian brokers offer varying levels of programmatic access:

Broker	API Type	Paper Trading	Asset Coverage	Latency
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Mirae Asset Sekuritas	REST + WebSocket	Yes	All IDX stocks	Medium
Phillip Sekuritas	REST API	Yes	All IDX stocks	Medium
Sinarmas Sekuritas	REST API	Limited	All IDX stocks	Medium
Ajaib / Stockbit	Partner API	Yes	Selected stocks	Low
Bareksa / Bibit	No direct API	N/A	Mutual funds only	N/A

4.1 API Requirements Checklist

- Real-time market data streaming (WebSocket preferred)
- Order placement: market, limit, stop-loss, stop-limit
- Order modification and cancellation
- Portfolio and position query
- Account balance and margin availability
- Transaction history and trade confirmations
- Paper/sandbox environment separate from production

5. Execution Engine

The execution layer translates detection signals into actual (paper) orders while managing risk and minimizing market impact.

5.1 Position Sizing Model

Fixed Fractional Risk: Risk maximum 1-2% of total paper capital per trade. Position size is calculated as: $Position\ Size = (Account\ Risk\ \%) / (Entry\ Price - Stop\ Loss\ Price)$. This ensures no single trade can disproportionately damage the portfolio.

5.2 Order Types & Execution Logic

Order Type	Use Case	IDX Support
Market Order	Immediate execution on strong signal	Yes
Limit Order	Control entry price, avoid slippage	Yes
Stop-Loss	Automatic exit at defined risk level	Yes
Stop-Limit	Stop-loss with price control	Yes
Iceberg / Hidden	Large orders without revealing size	Limited
Bracket Order	Entry + stop + target in one	Broker-dependent

5.3 Slippage & Liquidity Model

Indonesia's market has lower liquidity compared to developed markets. Slippage must be modeled realistically:

- **LQ45 stocks:** Assume 0.05-0.15% slippage for orders <IDR 500M
- **Mid-cap stocks:** Assume 0.2-0.5% slippage, scale position down
- **Small-cap / non-EMAS:** Avoid or use limit orders only; slippage can exceed 1%
- **Auction phases:** Higher volatility, model plusmn;0.3% additional slippage

6. Risk Management & Regulatory Compliance

Paper trading must still adhere to real-world constraints to produce valid backtesting and forward-testing results.

6.1 Automated Risk Controls

- **Maximum Daily Loss:** Halt trading if paper P&L drops >3% of capital
- **Single Stock Limit:** No more than 10% of portfolio in one ticker
- **Sector Concentration:** Max 30% exposure to any single sector
- **Correlation Check:** Avoid simultaneous entry into highly correlated pairs
- **Drawdown Circuit Breaker:** Pause system after 10% peak-to-trough drawdown

6.2 IDX & OJK Regulatory Constraints

- **Short Selling:** Heavily restricted; only permitted on select stocks with securities lending. System should default to long-only.
- **Auto-Rejection Thresholds:** Price movements beyond plusmn;10% (non-EMAS) or plusmn;20% (EMAS) trigger exchange halts. Orders outside these bands are rejected.
- **UPT (Unusual Price Movement):** Trading halt for 30+ minutes if price moves >10% without apparent cause.
- **Margin Requirements:** Minimum 50% for most stocks; system must check buying power before order submission.
- **Pre-Closing Session:** 15:45-16:00 WIB; orders can be placed but executed at closing auction price.

7. Indonesia Market-Specific Considerations

Understanding local market structure is critical for signal accuracy and realistic performance expectations.

7.1 Market Microstructure

- **Retail Dominance:** ~80% of accounts are retail; institutional flow is easier to isolate but retail FOMO creates noise after initial big money entry
- **Two-Session Trading:** 09:00-12:00 and 13:30-16:00 WIB; lunch break creates information gaps
- **Pre-Open & Closing Auctions:** 08:45-09:00 and 15:45-16:00; significant price discovery happens here
- **Lot Size:** 100 shares per lot; minimum order is 1 lot
- **Tick Size:** Varies by price band (IDR 1-25 depending on stock price)

7.2 Seasonal & Event Patterns

- **Dividend Season:** March-July; stocks often rally pre-ex-date then drop post-ex-date
- **Ramadan Effect:** Lower volume in fasting month; signals may be less reliable
- **Year-End Window Dressing:** Institutional portfolio rebalancing in December
- **Currency Volatility:** IDR/USD movements significantly impact foreign flow direction

7.3 Key Sectors & Concentration Risk

The IDX is heavily concentrated in banking (BBCA, BBRI, BMRI) and consumer sectors. Big money signals in these names may reflect index rebalancing rather than alpha-generating accumulation. The system should weight signals in non-benchmark names more heavily.

8. Recommended Technology Stack

Layer	Technology	Purpose
Language	Python 3.11+	Primary development language
Data Processing	Pandas, NumPy, Polars	Time-series analysis & signal computation
Database	PostgreSQL + TimescaleDB	Tick data storage & historical queries
Message Queue	Redis / RabbitMQ	Async signal distribution

Execution	Asyncio, aiohttp	Low-latency order submission
Backtesting	Backtrader / Zipline / Custom	Strategy validation on historical data
Monitoring	Grafana + Prometheus	Live P&L, signal dashboard, alerts
Deployment	Docker, AWS/GCP	Containerized, cloud-hosted infrastructure

9. Implementation Roadmap

A phased approach reduces risk and allows iterative validation of the core hypothesis: that big money flow is predictive of near-term price movement in IDX.

Phase 1: Foundation (Weeks 1-4)

1. Set up paper trading account with API-enabled broker
2. Build data ingestion pipeline for daily OHLCV + broker summary
3. Implement basic volume anomaly screener
4. Create manual alert system (email/Telegram) for high-score signals
5. Track alerts manually for 2-4 weeks to validate signal quality

Phase 2: Signal Refinement (Weeks 5-10)

1. Add real-time tick data ingestion
2. Implement composite BMS scoring algorithm
3. Backtest signals on 1-2 years of historical data
4. Add false signal filters (retail FOMO, news correlation)
5. Optimize signal weights using Sharpe ratio maximization

Phase 3: Automation (Weeks 11-16)

1. Connect to broker paper trading API
2. Build order execution engine with position sizing
3. Implement risk management circuit breakers
4. Deploy monitoring dashboard
5. Run fully automated paper trading for 1-3 months

Phase 4: Validation & Scaling (Weeks 17-24)

1. Analyze paper trading results vs. backtest expectations
2. Refine slippage and liquidity models based on real execution data
3. Add advanced features: multi-strategy, portfolio optimization
4. Prepare for potential live capital deployment (separate decision)

10. Critical Challenges & Mitigations

Challenge	Impact	Mitigation Strategy
Data Latency	Stale signals miss optimal entry	Use WebSocket feeds; colocate near exchange
Broker API Limits	Rate limiting blocks execution	Implement exponential backoff; queue orders
False Positive Signals	Retail noise mimics big money	Multi-factor confirmation; broker code analysis
Low Liquidity Slippage	Real fills worse than expected	Dynamic position sizing; limit order preference
Regulatory Changes	OJK rule changes break logic	Modular design; rapid rule engine updates
Data Provider Reliability	Feed outages cause missed signals	Redundant data sources; failover logic
Overfitting to History	Backtest looks great, live fails	Walk-forward analysis; out-of-sample testing

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